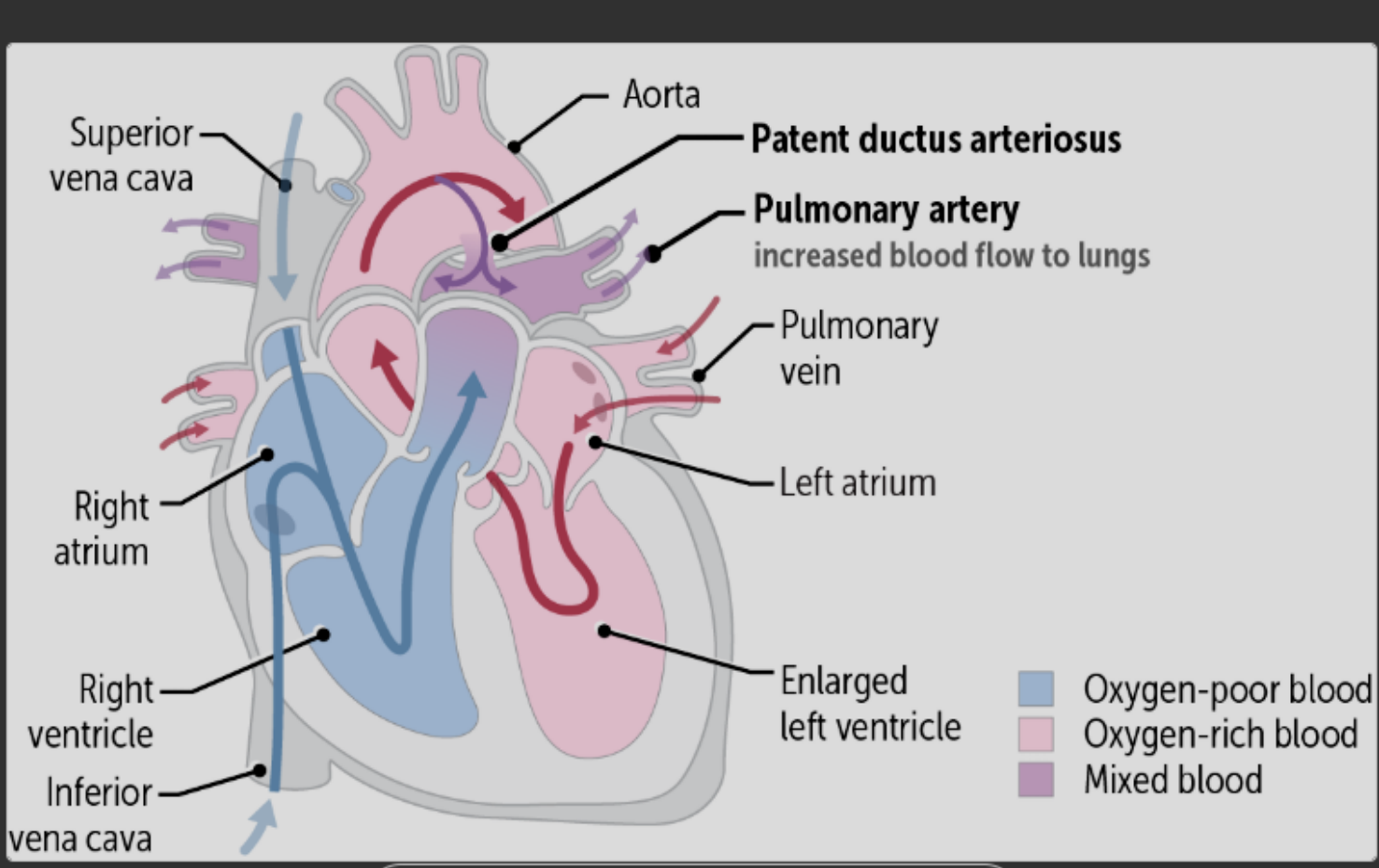
After birth, the bypass created by the DA is no longer needed. Once a newborn's lungs fill with air, the elevated oxygen concentration in the lung dramatically reduces resistance to blood flow. This decrease in pulmonary vasculature resistance facilitates blood flow through the pulmonary circulation, and the DA closes within 15-30 hours after birth. Closure of the DA is closely linked to declining blood levels of prostaglandin E1 (PGE_1).

Physiology

When the DA does not close as expected, it becomes a PDA (Figure 2).



new right-to-left shunt in the setting of CHD is known as Eisenmenger syndrome.

Such PDA patients may show **differential cyanosis** with only the lower extremities appearing blue. This occurs because in the right-to-left shunting, the deoxygenated blood will enter the aorta distal to where the arteries supplying the upper extremities exit the aorta (ie, left subclavian artery). So the upper extremities are supplied with normally oxygenated blood while blood supply to the lower extremities is reduced, causing lower extremity cyanosis (Figure 3).

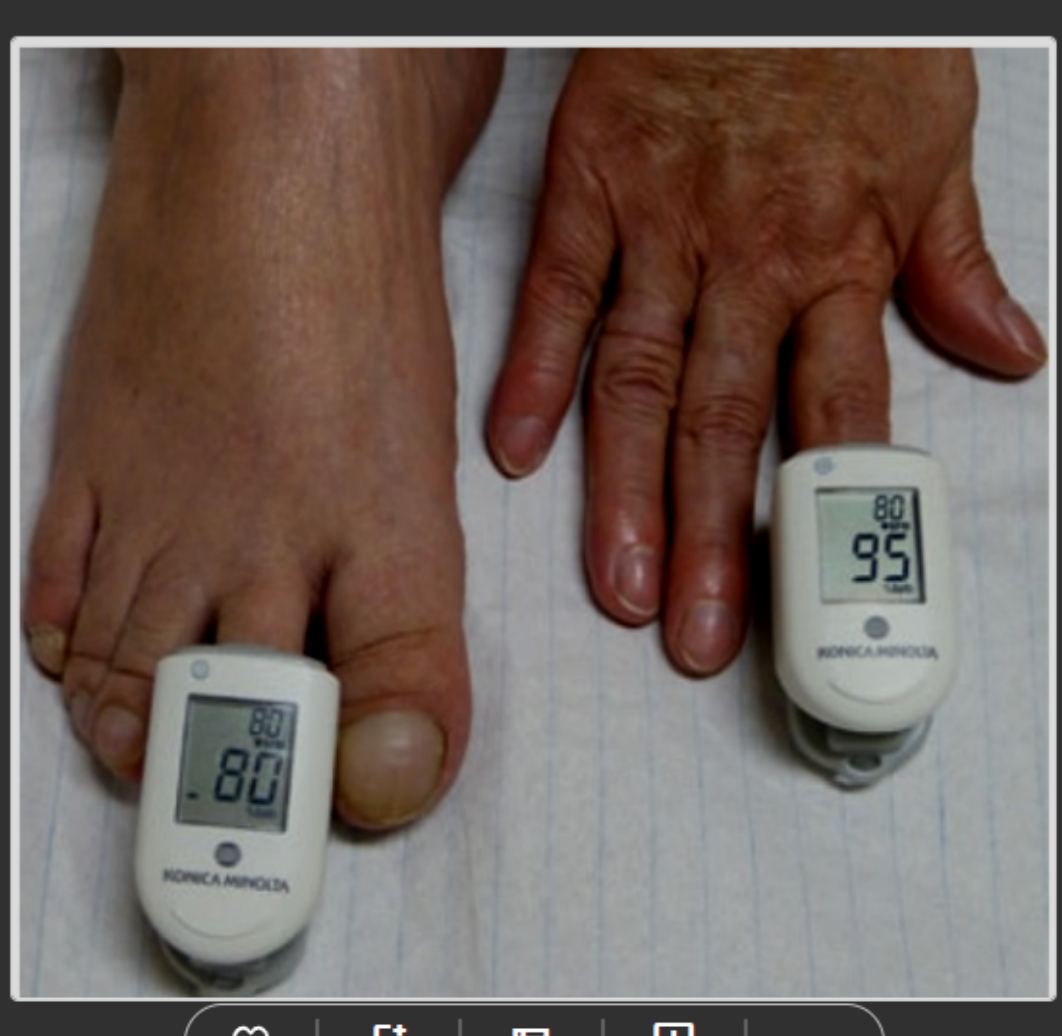
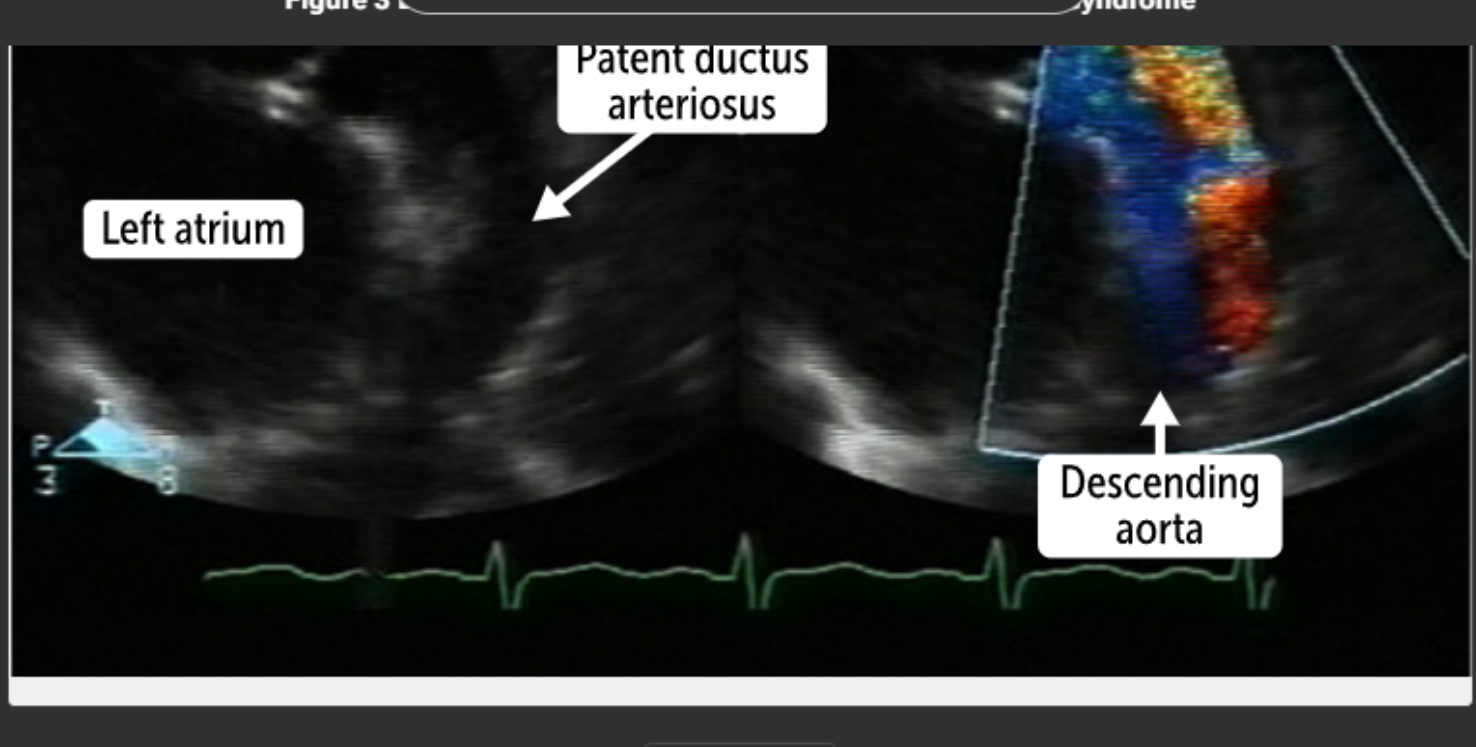


Figure 3 … yndrome



QUIZ

Figure 4 Echocardiography in patent ductus arteriosus

Doppler can detect the shunting of blood through the PDA and can be used to assess disease severity. Doppler flow studies quantify the flow through the PDA.

Occasionally, cardiac catheterization is used to measure changes in the oxygen saturation that arise because of cardiac defects. In a heart with a PDA, cardiac catheterization will demonstrate that the blood oxygen saturation in the pulmonary artery is higher than the blood oxygen saturation in the RV, because oxygenated blood is being shunted from the aorta into the pulmonary artery. (This elevation is known as a "step-up.")

Management

What Is Atrial Septal Defect? (LO#1)

An atrial septal defect (ASD) is a congenital heart defect in which an opening persists in the interatrial septum, the wall that normally separates oxygen-rich blood in the left atrium from the oxygen-poor blood in the right atrium. Because left atrial pressure is higher, blood moves through the defect from left to right, producing a **left-to-right shunt** (Figure 5).

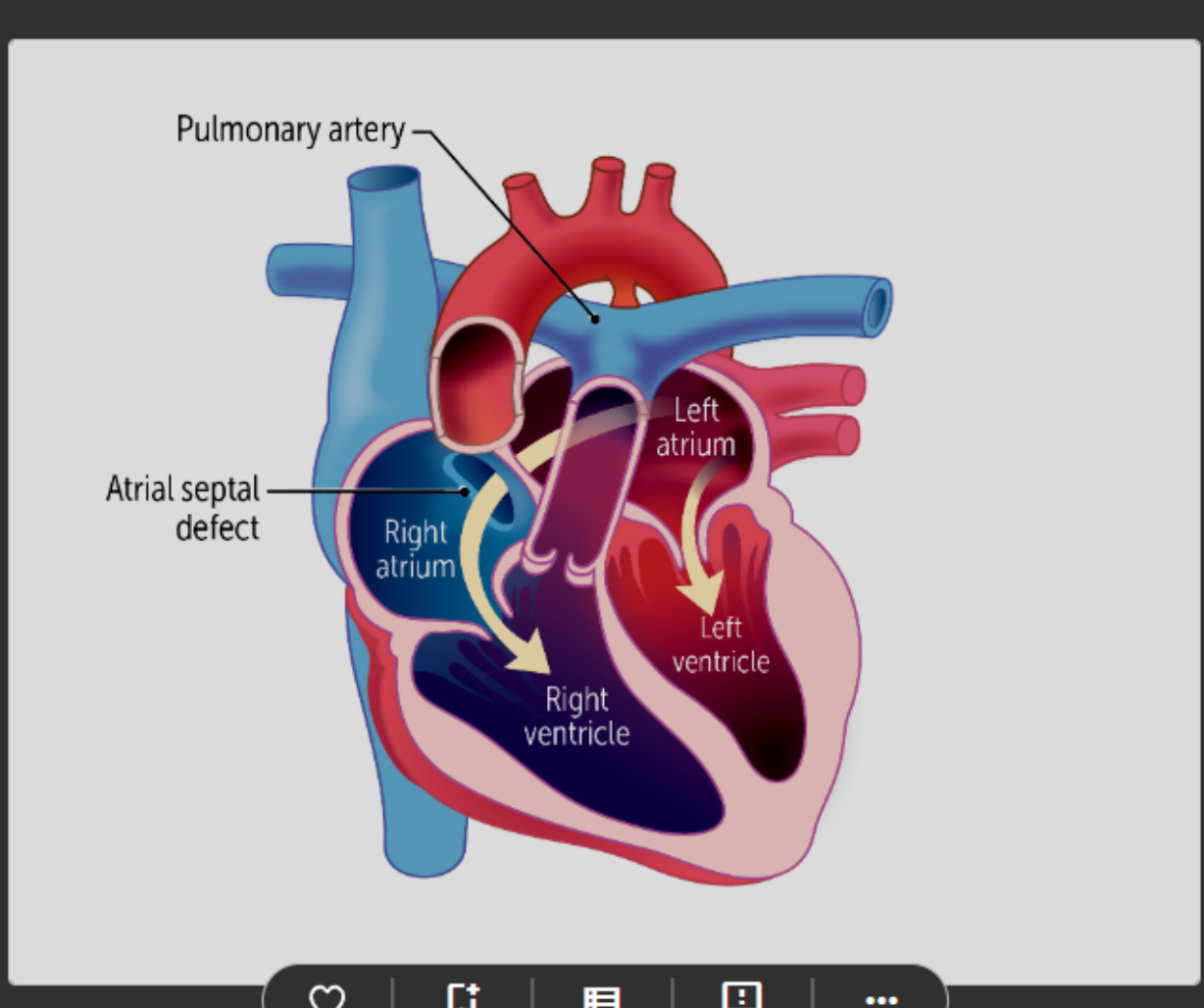


Figure 5 Atrial septal defect with resultant left-to-right shunt

Pathophysiology

An ASD is an abnormal communication between the RA and LA that persists after birth. To understand how this opening develops, it helps to recall the normal sequence of atrial septum formation during fetal life (Figure 6).

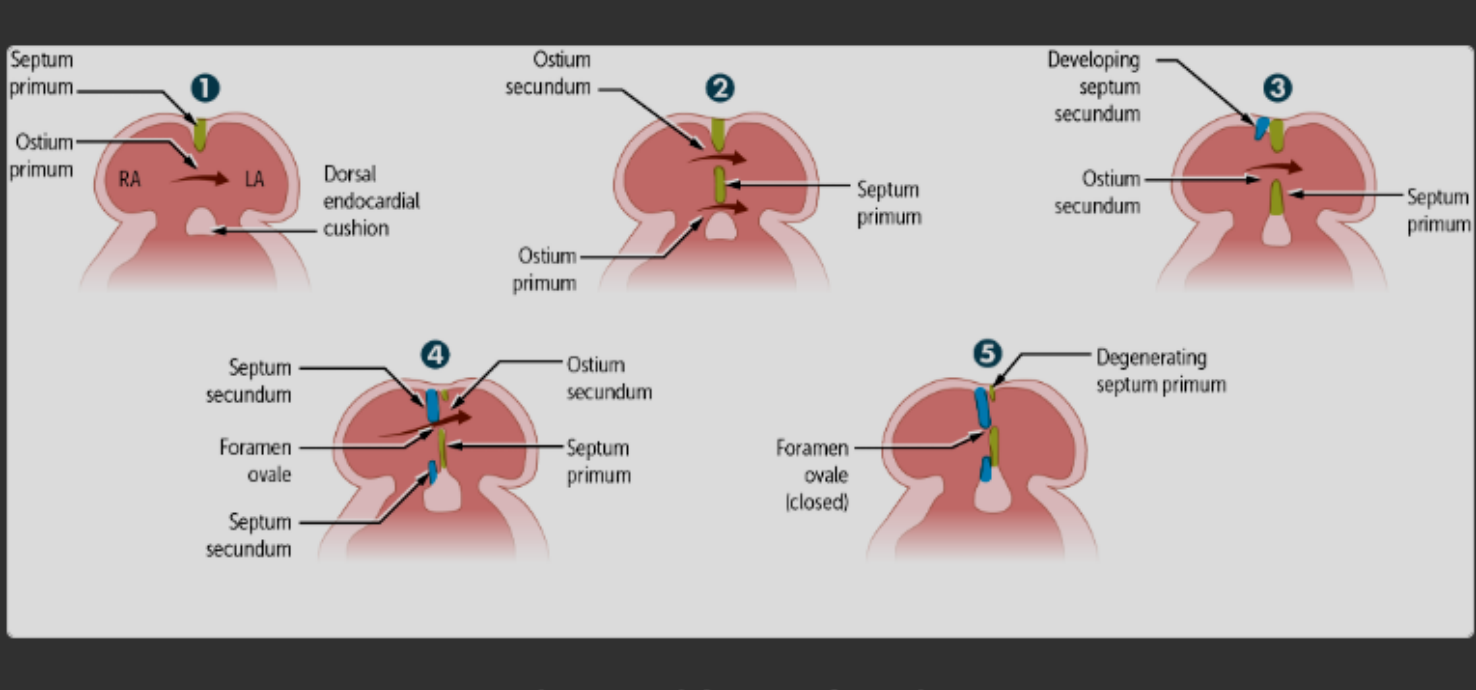


Figure 6 Atrial septum formation

The interatrial septum forms in two main steps:

1. **Septum primum formation** - A thin crescent of tissue grows downward from the roof of the primitive atrium toward the endocardial cushions, leaving a temporary opening called the **foramen primum**. As this gap closes, small perforations appear in the upper portion of septum primum that coalesce into the **foramen secundum**, allowing continued flow between the atria

Ostium Primum Defects. (15% to 20% of ASDs) Occur lower in the atrial wall when the septum primum fails to fuse with the endocardial cushions. This results in persistence of the foramen primum. These defects are generally located in the ventral portion of the atrial septum.

Ostium Secundum Defects. (70% of ASDs) Result from excessive resorption of the septum primum or inadequate growth of septum secundum. They occur near the fossa ovalis in the central portion of the atrial septum. Recall that the fossa ovalis is a depression in the atrial septum that marks where the **foramen ovale** closed after birth. The foramen ovale is an opening with a flap that allows blood flow from the right atrium to the left atrium, with the flap closing off after atrial systole.

> CLINICAL CORRELATION
>
> A patent foramen ovale (PFO)—failure of the foramen ovale to close after birth—is not considered a true ASD because no atrial septal tissue is missing. PFOs occur in up to 25% of the normal population, and most people with a PFO are asymptomatic.

Sinus Venosus and Coronary Sinus Defects. Although these defects do not arise in the atrial septum per se, they are functionally similar to defects in the atrial septum because they allow blood from the LA to be shunted into the RA. Sinus venosus defect … by a superior vena cava (SVC) or inferior vena cava (IVC) that abnormally enters the RA, … closure no longer varies with respiration (Figure 8).

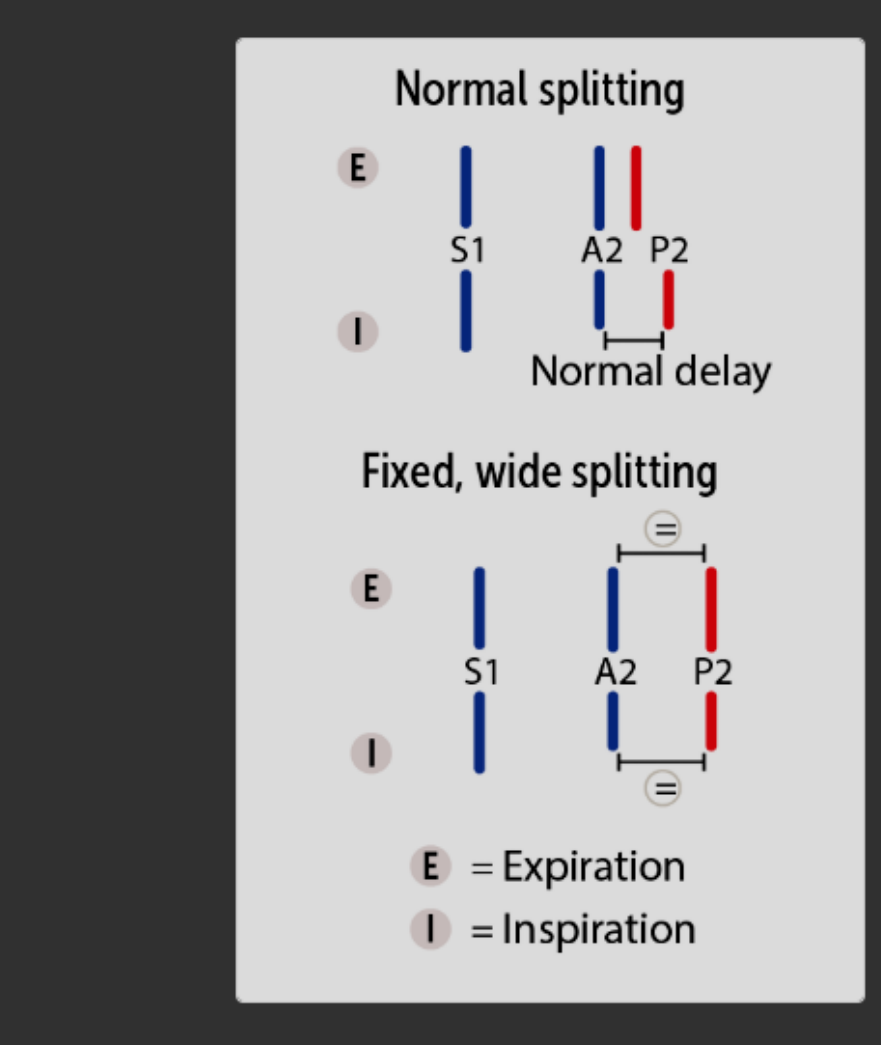


Figure 8: S2 splitting

The distinct nature of this heart sound is a reason why even asymptomatic ASDs can be detected on physical exam.

Interestingly, the ASD itself does not produce a murmur, since the pressure differences between the two atria is not significant enough to result in turbulent flow. However, a systolic flow murmur may be heard over the pulmonic valve in patients with large shunts, caused by increased flow across the valve. Pulmonic … murmur in the second left intercostal space radiating to the back (please see video). It

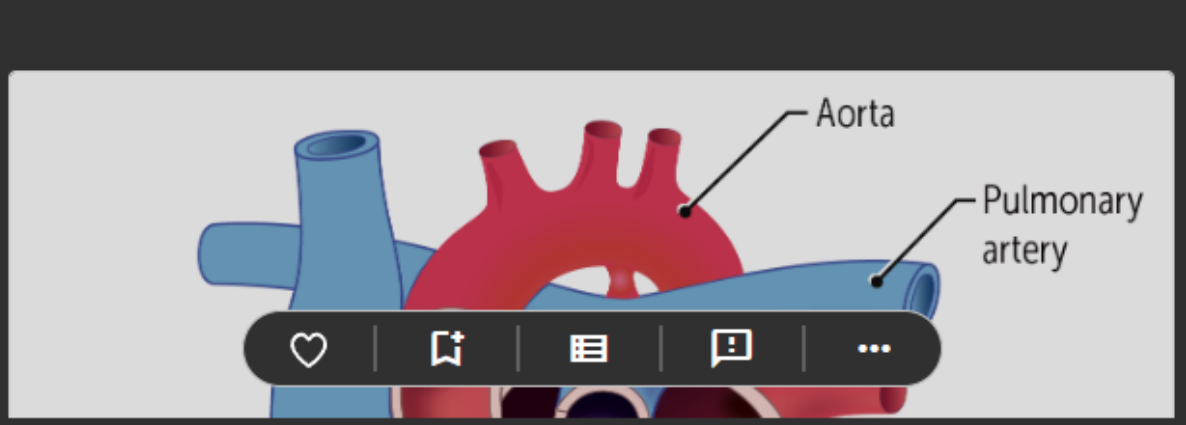
Management

For small ASDs, corrective intervention may be unnecessary, as spontaneous closure may occur in young patients. Defects less than 8 mm in diameter close on their own in nearly 80% by 15 months of age. The likelihood of spontaneous closure decreases with age, so persistent small ASDs that cause no significant shunting are usually monitored over time to ensure that continued non-intervention remains appropriate.

Larger defects—or small defects with a large amount of shunting—should be repaired even in the absence of symptoms. Interventions for repair include surgery or percutaneous closure. Percutaneous repair is now generally the procedure of choice because it is less invasive.

What Is Ventricular Septal Defect? (LO#1)

Ventricular septal defect (VSD) is the most commonly diagnosed congenital cardiac anomaly in infancy, occurring in 4 of every 1000 births. As the name suggests, a VSD is caused by an abnormal opening in the tissue of the interventricular septum, which normally separates the highly oxygenated blood of the left ventricle (LV) from the poorly oxygenated blood of the right ventricle (RV), as shown in Figure 9.



ALICE L WALTON SCHOOL OF MEDICINE

Obstructive Congenital Heart Defects

Last updated January 13, 2026 17 min read ScholarRx
Adapted by: Alexis Gillett, Chelsey Deel, Stephen Nix, Rhome Hughes
Editors: Russell Johnson, Jordan Finn
Contributor: Nancy Halliday
Reviewers: Diptiman Bose, Carrie Elzie
Adapted from: Acyanotic Obstructive Congenital Heart Defects

Listen to this Brick
0:00 / 26:55 1x

Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of obstructive congenital heart defects.

What do you want to learn?

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Learning Objectives (1) Versions

After completing this brick, you will be able to:

1 Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of obstructive congenital heart defects.

exact mechanism of either type is not fully understood.

Acquired CoAs are rare, but they arise in adulthood because of severe atherosclerosis or inflammatory diseases of the aorta, such as Takayasu arteritis.

From what embryonic structure does the arch of the aorta arise?

Physiology. Because CoA is an obstructive lesion, the main hemodynamic issue is increased resistance to blood flow through the narrowed segment of the descending aorta. This is problematic because tissues distal to the coarctation require oxygen-rich blood to survive. So, the body adapts in at least one of two ways to get blood past the narrowed aorta:

- By increasing systemic blood pressure: higher blood pressures proximal to the aortic narrowing will increase arterial pressure proximal to the stenosis (eg, in the arms … ng cause of

How does the body compensate for increased resistance to blood flow in CoA?

Complications

If left untreated, patients can develop a variety of complications, all of which are related to the secondary hypertension that arises from CoA. These include **left ventricular hypertrophy (LVH)** from increased LV afterload, atherosclerosis, and cerebral aneurysms.

What are common complications associated with CoA?